

Research experience

- 2024 – ... **Lecturer**, University of Bristol (UK)
- 2024 – 32 **Royal Society Dorothy Hodgkin Fellow**, University of Bristol (UK)
- 2022 – 24 **Research associate**, University of Bristol (UK), PI: Jens Marklof
Researcher Co-Investigator on EPSRC Standard grant EP/W007010/1.
- 2021 – 22 **Postdoctoral fellow**, Max Planck Institute for Mathematics, Bonn (DE), mentored by Ursula Hamenstädt

Education

- 2018 – 21 **PhD in Mathematics** under the supervision of Nalini Anantharaman, Université de Strasbourg (FR)
Geometry and spectrum of typical hyperbolic surfaces
- 2014 – 18 **École Normale Supérieure** in Mathematics with minor in Physics, Paris
- 2017 – 18 **Master's degree in Mathematics** (with honours), Paris-Sud University
- 2017 **Agrégation of Mathematics** (7th place)
Secondary-school and University teaching degree
- 2014 – 15 **Bachelor's degree in Mathematics** (with honours), Paris-Sud University
- 2014 – 15 **Bachelor's degree in Physics** (with honours), Paris-Sud University
- 2012 – 14 **Preparatory class**, Lycée Privé Sainte Geneviève, Versailles (FR)

Articles

Publications in peer-reviewed journals

- (6) Marklof J., Monk L., *The moduli space of twisted Laplacians and random matrix theory*, International Mathematics Research Notices (2024), <https://doi.org/10.1093/imrn/rnae239>.
- (5) Monk L., Stan R., *Spectral convergence of the Dirac operator on typical hyperbolic surfaces of high genus*, Annales Henri Poincaré (2024), <https://doi.org/10.1007/s00023-024-01452-z>.
- (4) Anantharaman N., Monk L., *A high-genus asymptotic expansion of Weil-Petersson volume polynomials*, Journal of Mathematical Physics, 63, 043502 (2022).
- (3) Monk L., Thomas J., *The tangle-free hypothesis on random hyperbolic surfaces*, International Mathematics Research Notices, Volume 2022, Issue 22, November 2022, Pages 18154–18185.
- (2) Monk L., *Benjamini-Schramm convergence and spectrum of random hyperbolic surfaces of high genus*, Analysis & PDE, Vol. 15 (2022), No. 3, 727–752.
- (1) Fouvry J.B., Pichon C., Chavanis P., Monk L., *Resonant thickening of self-gravitating discs: imposed or self-induced orbital diffusion in the tightly wound limit*, Monthly Notices of the Royal Astronomical Society, Volume 471, Issue 3, November 2017, Pages 2642–2673.

Preprints

- 2025 Anantharaman N., Monk L., *Friedman-Ramanujan functions in random hyperbolic geometry and application to spectral gaps II*, arxiv:2502.12268.
- 2024 Anantharaman N., Monk L., *Spectral gap of random hyperbolic surfaces*, arxiv:2403.12576.
- 2024 Anantharaman N., Monk L., *A Moebius inversion formula to discard tangled hyperbolic surfaces*, arXiv:2401.01601.
- 2023 Anantharaman N., Monk L., *Friedman-Ramanujan functions in random hyperbolic geometry and application to spectral gaps I*, arXiv:2304.02678.

Awards and grants

- 2024 – 32 Royal Society Dorothy Hodgkin Fellowship.
- 2024 Maryam Mirzakhani New Frontiers Prize.
- 2022 – 25 Researcher Co-Investigator on EPSRC Standard Grant EP/W007010/1.
- 2022 Best PhD in Mathematics prize from the University of Strasbourg.
- 2021 Prize L'Oréal-UNESCO Young Talents France for Women in Science.
- 2019 Best poster in Mathematics prize from the Doctoral School.
- 2018 – 21 PhD grant from the ENS (Paris, FR).
- 2014 – 18 Full scholarship at the ENS (Paris, FR).

Talks

Invited talks at conferences

- Sept 2026 Plenary talk at the General Meeting of the European Women in Mathematics, *University of Warwick (UK)*.
- July 2026 Special Joint Sectional Lecture with Frédéric Naud at the International Congress of Mathematicians, *Philadelphia (US)*.
- May 2026 Random and arithmetic models in spectral theory, *Isaac Newton Institute, Cambridge (UK)*.
- Nov 2025 Supporting Lecture to Jens Marklof's Presidential Address at the LMS Annual General Meeting, *London (UK)*.
- Nov 2025 Journées de dynamique, *Paris (FR)*.
- Aug 2025 LMS Global Meeting, *Hanoi (VN)*.
- Apr 2025 Probability, Analysis and Dynamics, *Bristol (UK)*.
- Apr 2025 Current Development of Mathematics Conference, *MIT-Harvard (US)*.
- Nov 2024 Random hyperbolic surfaces and random graphs, *CIRM, Marseille (FR)*.
- Dec 2022 Brunel-Bielefeld Workshop on Random Matrix Theory and Applications, *London (UK)*.
- Sept 2022 Shape Optimisation and Geometric Spectral Theory, *Edinburgh (UK)*.
- July 2022 ANR workshop "Quantum Chaos, Randomness and Spectral Problems in Mathematical Physics", *Paris (FR)*.
- July 2022 Journées Nancéennes de Géométrie, *Nancy (FR)*.
- May 2022 Laplacians on random hyperbolic surfaces and on random graphs, *Northwestern University (US)*.
- Feb 2022 14th Meeting of the GdR Quantum Dynamics, *Toulouse (FR)*.
- Feb 2022 Spectra and Dynamics on (Locally) Symmetric Spaces, *Paderborn (DE)*.
- Nov 2021 ANR "Aléatoire, Dynamique et Spectre" meeting, *Nantes (FR)*.

- Sept 2021 Hyperbolic dynamical systems and resonances, *Porquerolles (FR)*.
- Dec 2020 Jeunes géomètres dynamiques, *GdR Platon, Rennes (FR)*.
- Oct 2020 Billiards and Surfaces à la Teichmüller and Riemann Online.
- Nov 2019 8th Strasbourg / Zurich Meeting: Frontiers in Analysis and Probability, *Zurich (CH)*.
- May 2019 Spectral Theory and pRObability in Mathematical physics, *Strasbourg (FR)*.
- Apr 2019 Dynamics of geodesic flows and applications to PDEs, *Palaiseau (FR)*.

Colloquia

- Oct 2025 University of Warwick (UK).
- June 2025 Morning speaker at British Mathematical Colloquium, Exeter (UK).
- Nov 2024 University College London (UK).
- Sept 2024 University of Bristol (UK).
- March 2022 Dalhousie University (CA, online).

Mini-courses

- March 2023 GEoMetric Spectral Theory Online NEtwork minicourse, *Centre de Recherches Mathématiques (Montréal, CA)*.
- Oct 2022 Mini-course at the summer school “Random walks and related random topics”, *University of Göttingen (DE)*.
- May 2022 Advance courses “Spectrum of the Laplacian and Geometry”, *University Cergy-Pontoise (FR)*.

Seminars

- 2025 Geometry and Topology Colloquium, *Cambridge (UK)*. London Analysis Seminar, *London (UK)*. Geometric Group Theory + Random Matrix Theory, *Oxford (UK)*. Analysis and Geometry, *Rennes (FR)*. Geometry and Dynamics, *Lille (FR)*.
- 2023-24 [No travel due to pregnancy and 1 year-long maternity leave.]
Analysis, *Bristol (UK)*. Mathematical Physics, *Sussex (UK)*. Topology and Dynamics, *Birmingham (UK)*. Spectral Geometry in the Clouds x2 (*online*). Geometry and Topology, *Bristol (UK)*. Munich–Copenhagen–Santiago Seminar in Mathematical Physics (*online*).
- 2022 London Analysis seminar, *London (UK)*. Spectra/Moduli seminar, *Durham (UK)*. Differential Geometry and Topology, *Cambridge (UK)*. Analysis and Geometry, *Bristol (UK)*. Reading seminar, *Paris Nord (FR)*. Topology and Dynamics, *Orsay (FR)*.
- 2021 Ergodic Theory and Dynamical Systems, *Bristol (UK)*. Probability and Statistics, *Marseille (FR)*. Geometry and Topology, *Jussieu (FR)*. Algebra, Geometry and Physics, *Bonn (DE)*. Young researchers in spectral geometry II, *Spectral geometry in the clouds (online)*. Number theory, *Tel-Aviv (IL)*. Probability and Mathematical Statistics, *Milan–Pavia (IT)*. GdR Platon (*online*). Geometry and Topology, *Toulouse (FR)*. Dynamical systems and ergodic theory, *Zurich (CH)*.
- 2020 Analysis, *Cardiff (UK)*. Geometry, *Leeds (UK)*. Dynamical systems, *Bremen–Oldenburg (DE)*. Oberseminar Differentialgeometrie, *Bonn (DE)*. Probabilities, *Orsay (FR)*. Geometrical Analysis by Frédéric Naud (*online*).
- 2019 Spectral theory and Geometry, *Grenoble (FR)*.
- 2018 Analysis, *Strasbourg (FR)*.

Teaching & student supervision

Teaching

- Spring 25 **Tutorials**, University of Bristol
- Jan–Feb 21 **Substitute Lecturer** (2nd year analysis), University of Strasbourg
- 2018 – 21 **Teaching assistant**, University of Strasbourg:
2nd and 3rd year exercise sessions, preparation of the agrégation, math club for high-school students, 1st and 2nd year oral exams.
- 2018 **Higher-education teaching degree** (“agrégation”)
Competition granting a life-long position as a secondary-school or higher-education teacher, ranked 7th out of 315 admitted (1545 candidates).
- 2014 – 18 **Oral examiner** in preparatory schools (1st and 2nd year)

PhD students

- 2024 – 28 Daniel Meriaz, joint with Jens Marklof.
- 2025 – 29 Aapo Karvonen, joint with Jens Marklof.

Project students

- 2025 – 26 Jules Duhamel (ENS Paris), random circle packings.
- 2025 – 26 Adhyayan Sharma (University of Bristol), Lean project in analysis.
- Spring 23 Supervision of PhD Student Rares Stan (Institute of Mathematics of the Romanian Academy) on one research project (leading to article (5)).

Organisation of events

- June 2026 CMI-HIMR Summer School on Random Geometries and Random Matrices
- 2025 – ... **Creator and organiser of the Mathematical Explain & Explore Talks** at the University of Bristol (internal colloquium series).
- 2025 – ... **Analysis and Geometry seminar** in the University of Bristol, with Michiel van den Berg, Asma Hassannezhad, John Mackay and Safoura Zadeh.
- 2021 – ... **Online seminar “Spectral Geometry in the Clouds”** with Alexandre Girouard and Jean Lagacé.
- 2022 – 23 **Ergodic Theory and Dynamical Systems seminar** in the University of Bristol, with Thomas Jordan and Henna Koivusalo.
- 2019 **Organiser of reading group** on Mirzakhani’s work on random hyperbolic surfaces in Strasbourg, with Yohann Bouilly and Federica Fanoni.

Outreach activities

- 2025 & 26 **Panelist**, *Women and Girls in STEM event*, University of Bristol
- May 2025 **Public Lecture**, *Celebrating Women in Mathematics*, Iranian Mathematical Society (IR, online)
- 2022 **Outreach talk**, program “Access to Bristol”
- 2018 and 19 **Organisation of the RJM**, *University of Strasbourg (FR)*
Three-day meeting for high school girls to conduct mathematical group activities, meet female researchers and engineers and speak about gender issues.
- 2017 and 21 **Jury for the TFJM²**
Conference-like mathematical competition for high-school students.
- 2017 **Outreach talk**, *Lycée Privé Sainte-Geneviève (FR)*